

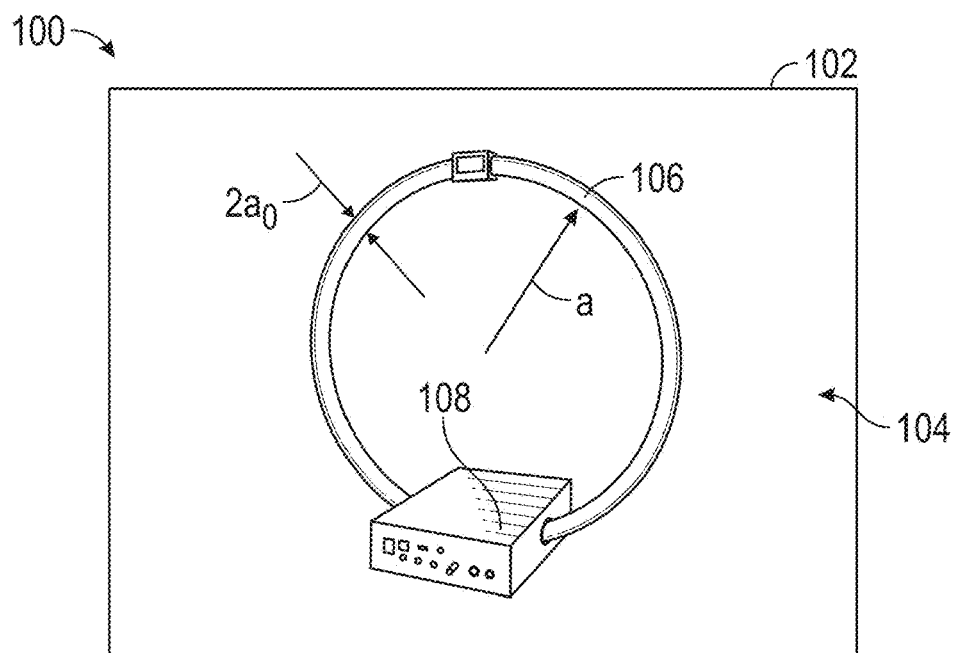


FIG. 1

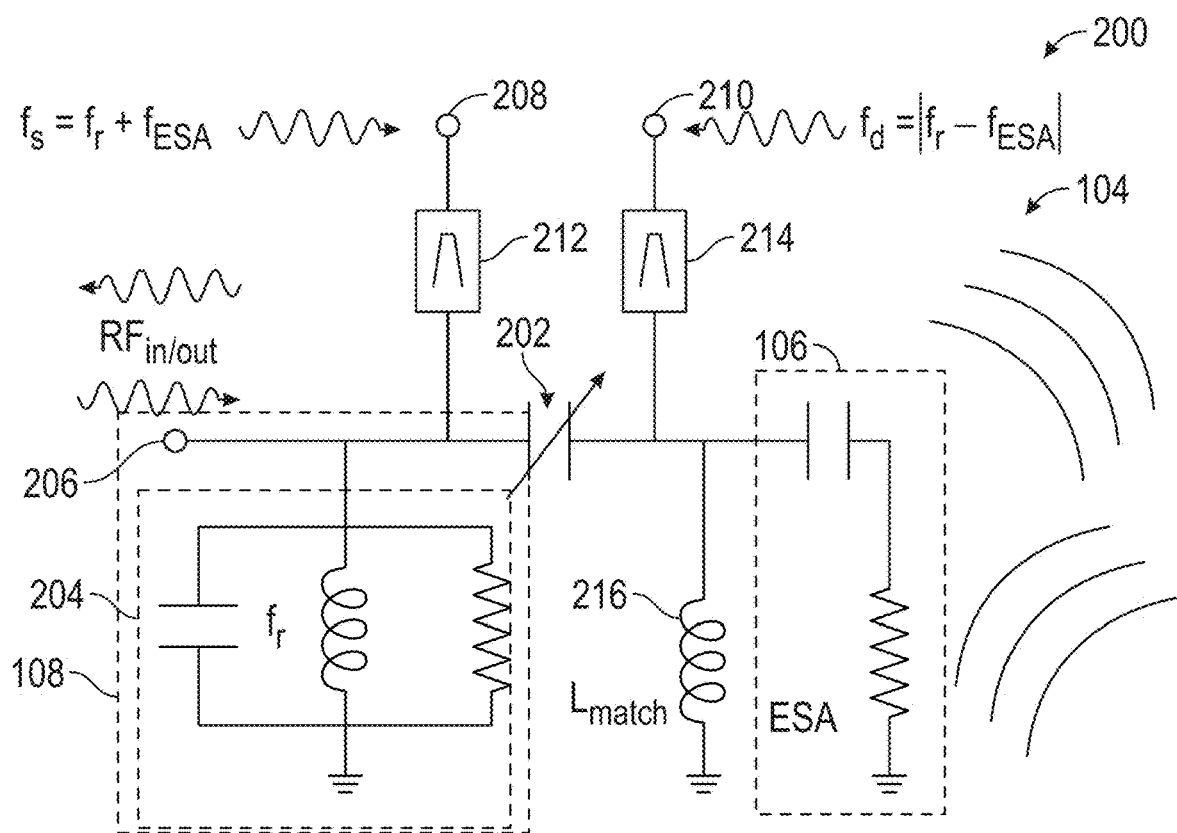
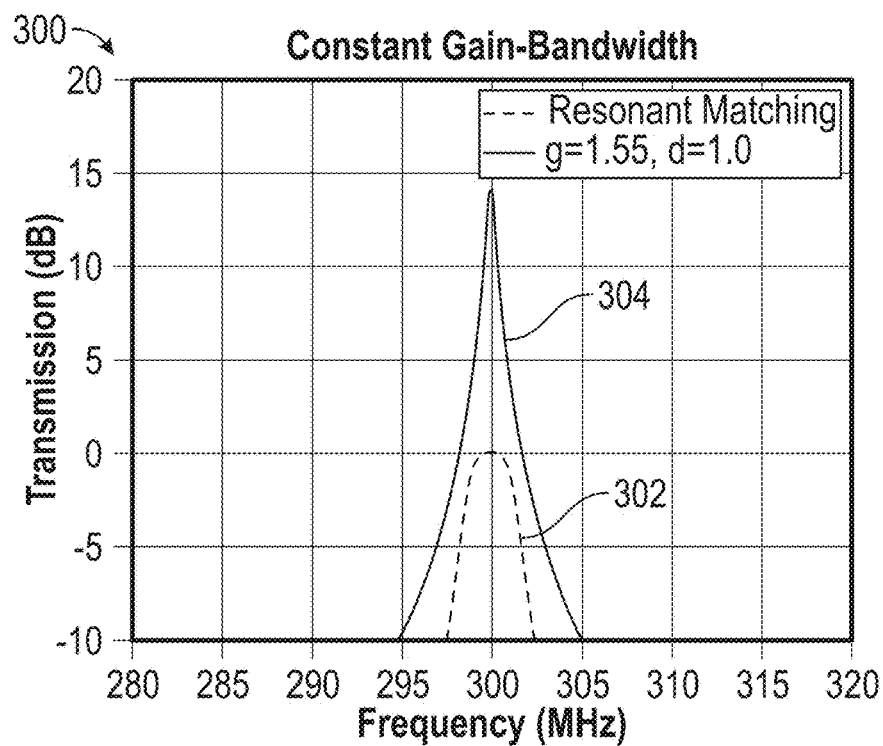
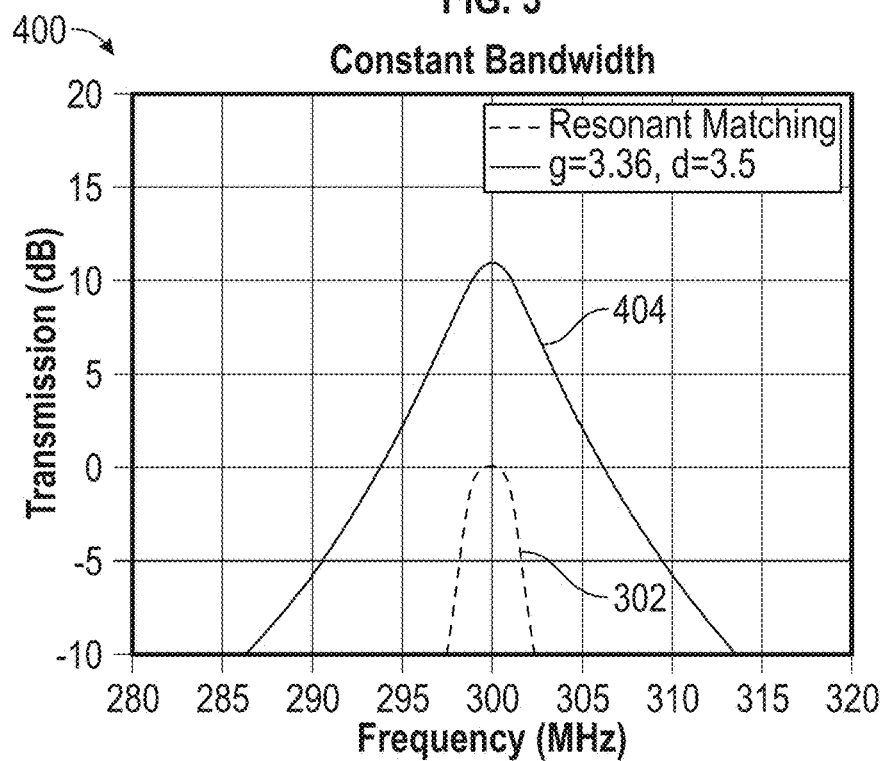


FIG. 2



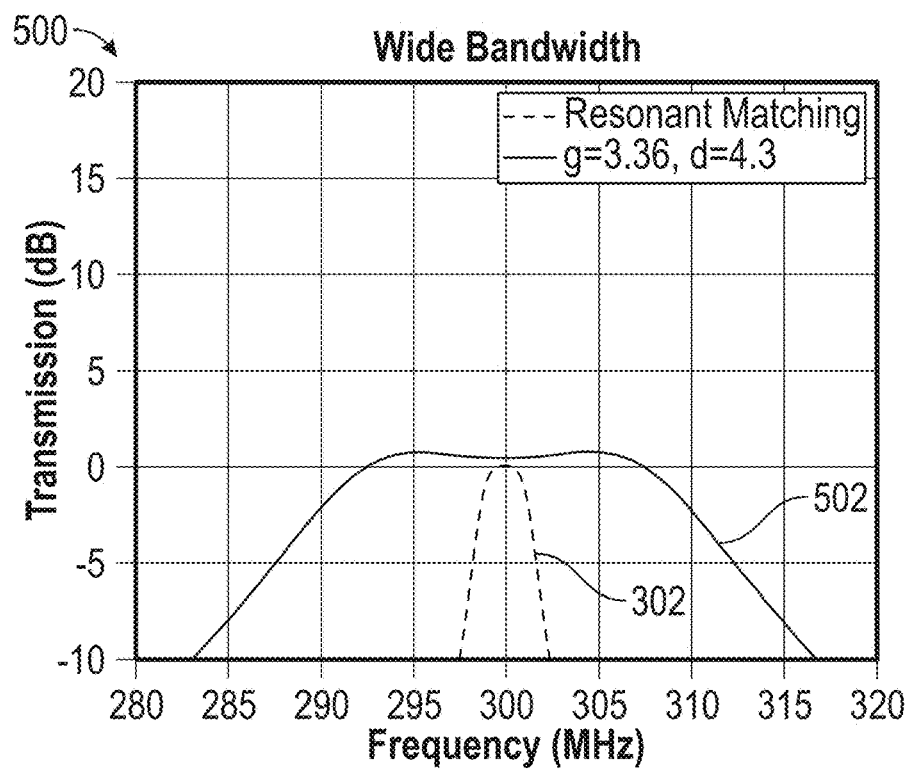
$$d^2 - g^2 < \frac{K_r K_{ESA}}{4}$$

FIG. 3



$$d^2 - g^2 = \frac{K_r K_{ESA}}{4}$$

FIG. 4



$$d^2 - g^2 > \frac{K_r K_{ESA}}{4}$$

FIG. 5

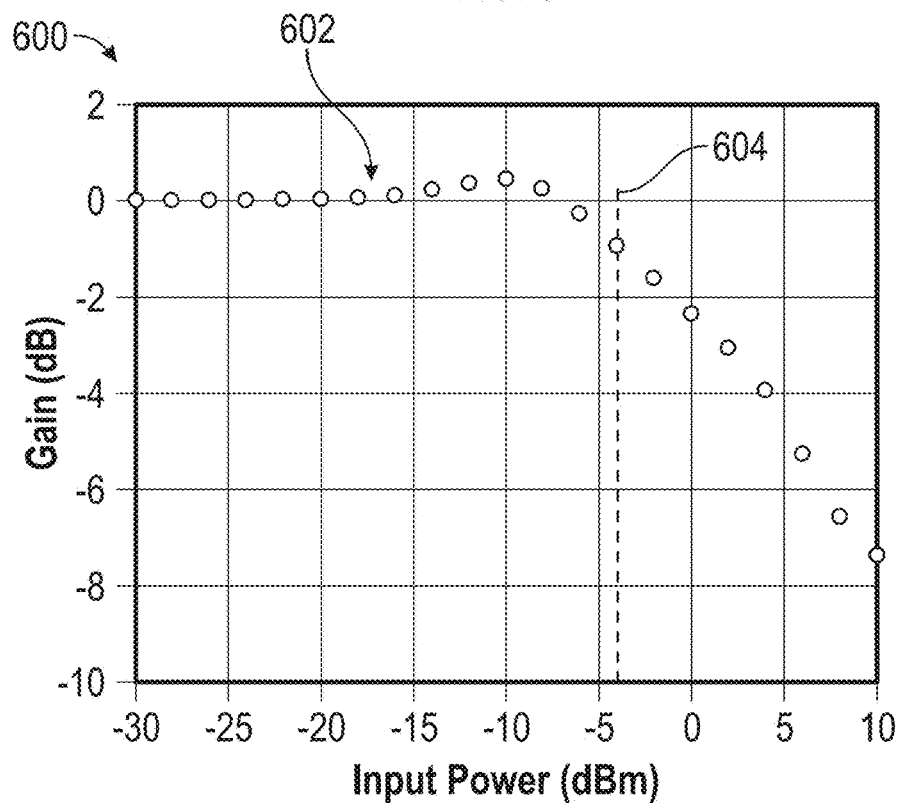
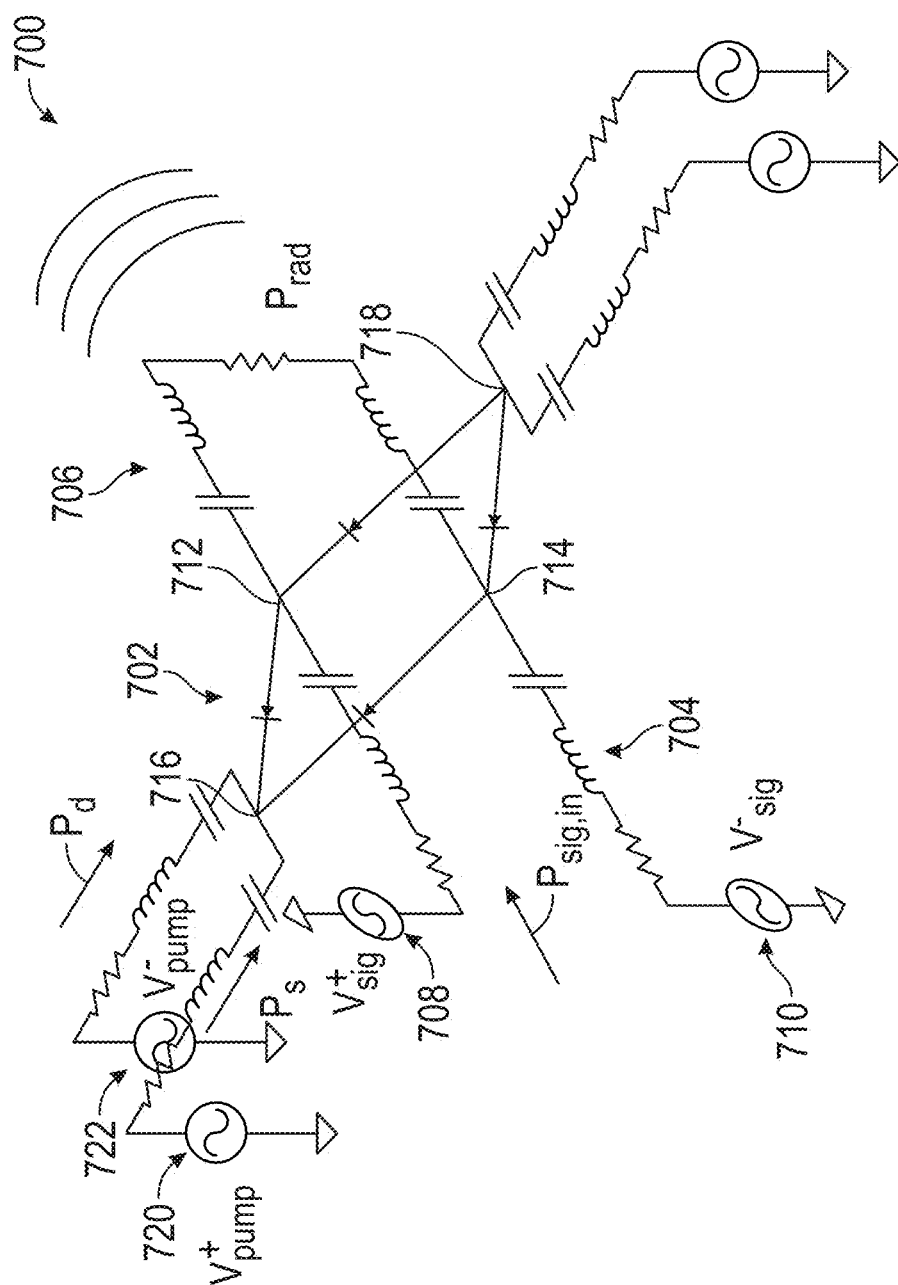


FIG. 6



7
G
L

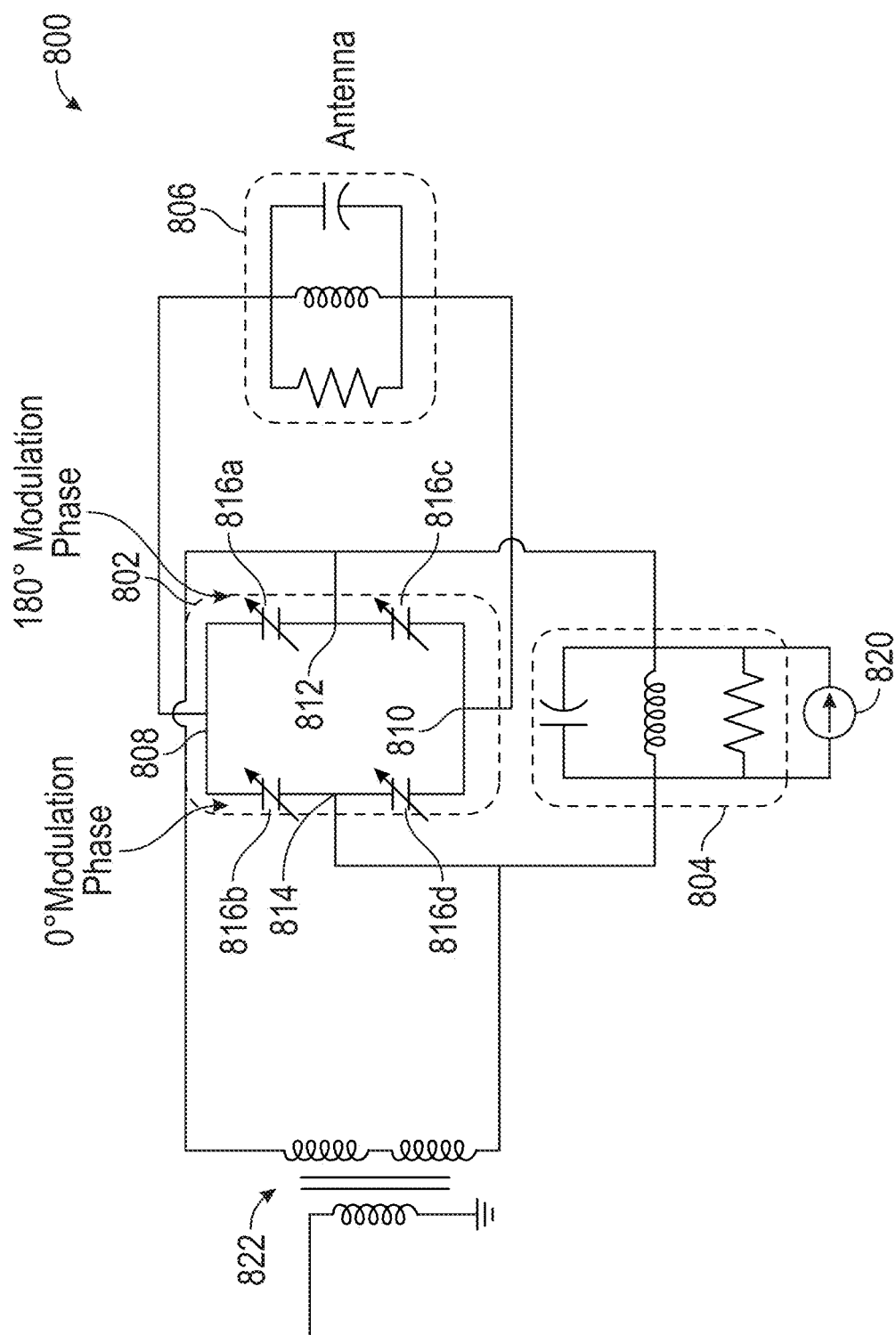


FIG. 8

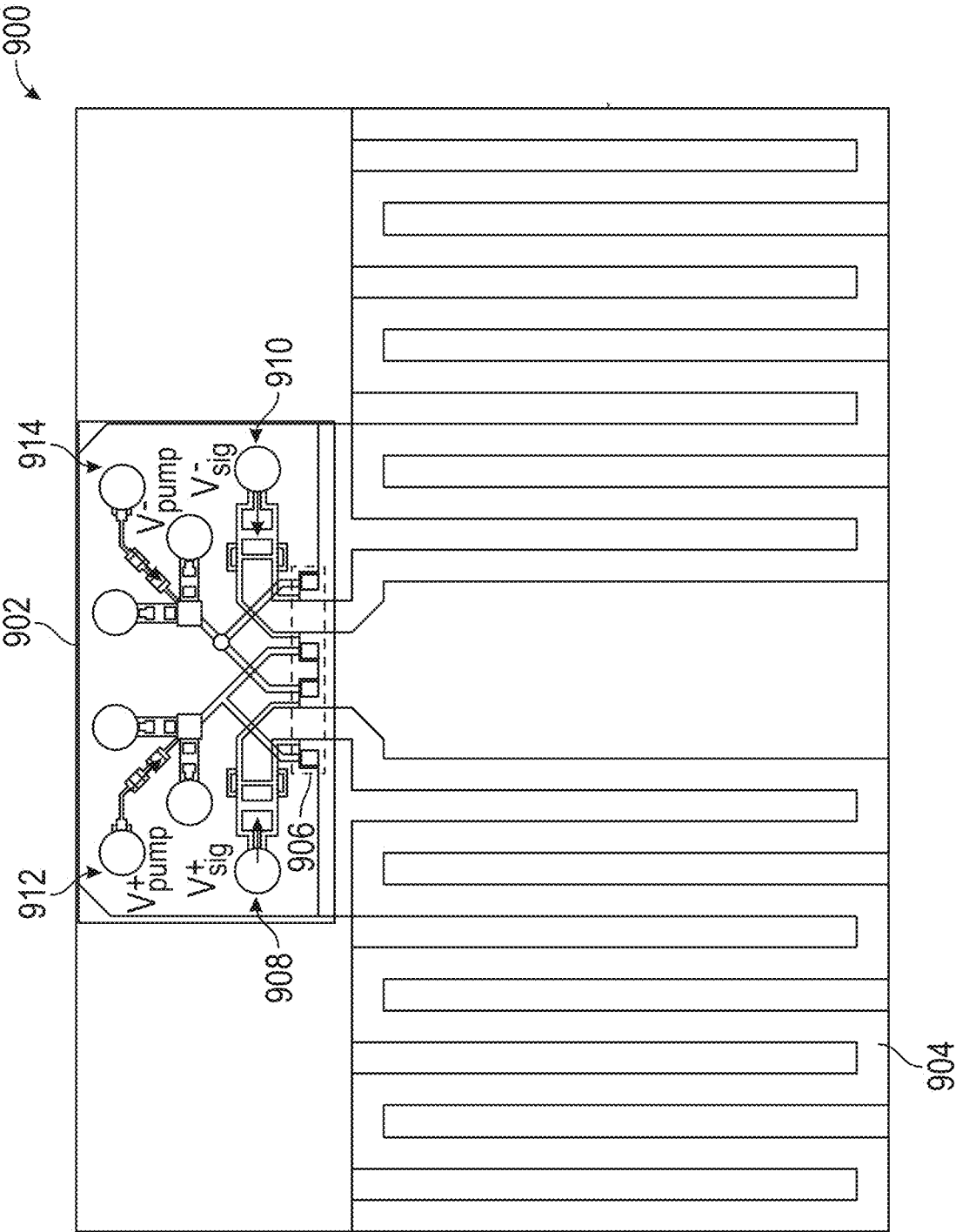


FIG. 9

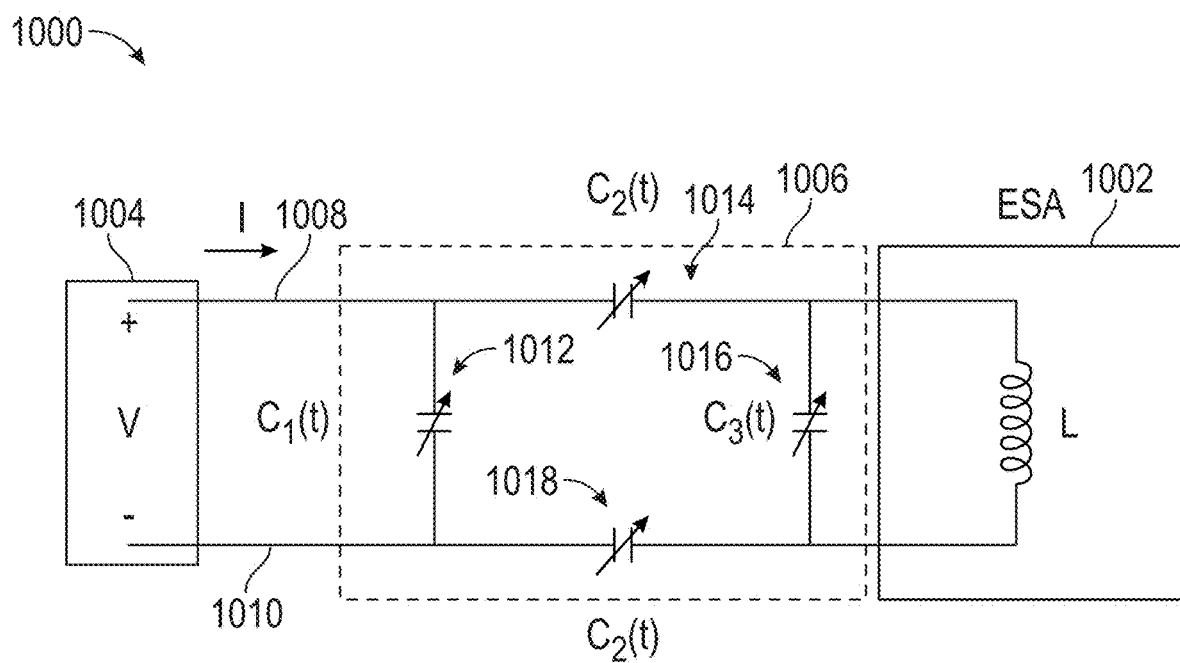


FIG. 10

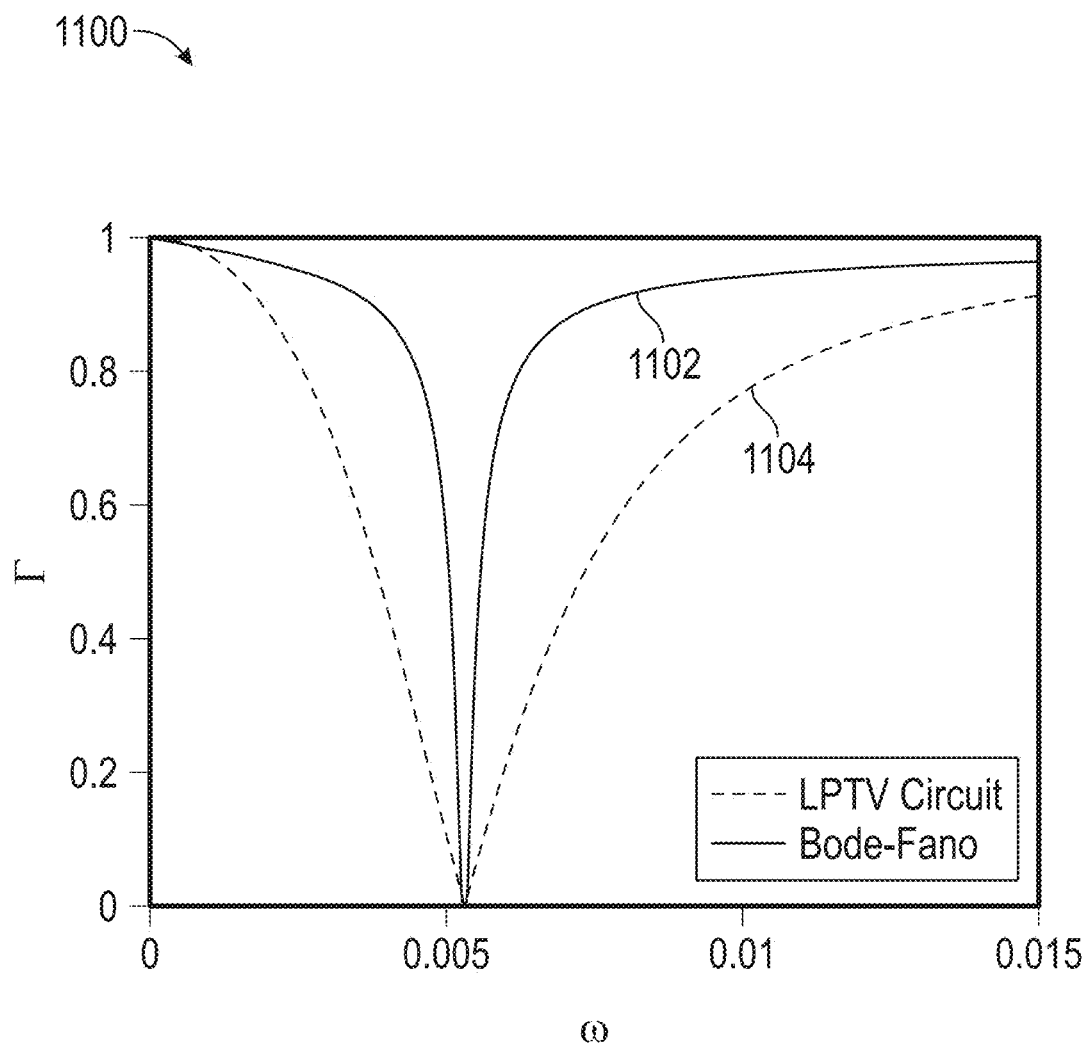


FIG. 11

MULTI-TONE MODULATED ANTENNA

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of U.S. Provisional Application No. 63/647,215 filed May 14, 2024, the disclosure of which is incorporated herein by reference in its entirety.

GOVERNMENT LICENSE RIGHTS

[0002] This disclosure was made with Government support under N66001-22-C-4501 awarded by IARPA, Intelligence Advanced Research Projects Activity. The Government has certain rights in the disclosure.

BACKGROUND

[0003] The present disclosure relates to electrically small antennas (ESAs) and, more specifically, to a system and method for operating an ESA as a dual tone modulated antenna.

[0004] Electrically small antennas (ESAs) are significantly shorter than the wavelength of the signals it is designed for. An ESA typically takes the form of a small loop or patch and is therefore advantageous to use when space is at a premium. Due to the fundamental tradeoff between antenna size (i.e., antenna radius), antenna wavelength and antenna bandwidth, ESAs are typically narrow band antennas. This tradeoff limits a communication data rate for the ESA, especially in a range of high frequency (HF) (about 30 MHz to about 300 MHz) and very high frequency (VHF) (about 300 MHz to about 3000 MHz). For various uses, it is desirable to have antennas as small as possible. Therefore, there is a need to operate an ESA that maintains a suitable communication bandwidth and efficiency.

SUMMARY

[0005] According to one embodiment, an antenna assembly is disclosed. The antenna assembly includes an electrically small antenna (ESA), a drive circuit for operating the ESA, the drive circuit including a resonator having a first resonant frequency, a variable coupling element that controls an electrical connection between the drive circuit and the ESA, a first modulator configured to modulate the variable coupling element at a first tonal frequency, wherein the first tonal frequency is a sum of the first resonant frequency and a second resonant frequency of the ESA, and a second modulator configured to modulate the variable coupling element at a second tonal frequency, wherein the second tonal frequency is a difference between the first resonant frequency and the second resonant frequency.

[0006] According to another embodiment, a device is disclosed. The device includes an electrically small antenna (ESA), a drive circuit for operating the ESA, the drive circuit including a resonator having a first resonant frequency, a variable coupling element that controls an electrical connection between the drive circuit and the ESA, a first modulator configured to modulate the variable coupling element at a first tonal frequency, wherein the first tonal frequency is a sum of the first resonant frequency and a second resonant frequency of the ESA, and a second modulator configured to modulate the variable coupling element at a second tonal frequency, wherein the second tonal

frequency is a difference between the first resonant frequency and the second resonant frequency.

[0007] According to another embodiment, a method of operating an electrically small antenna is disclosed. The method includes controlling an electrical connection between the electrically small antenna and a drive circuit via a variable coupling element, the drive circuit including a resonator having a first resonant frequency. Controlling the electrical connection includes modulating the variable coupling element at a first tonal frequency via a first modulator, wherein the first tonal frequency is a sum of the first resonant frequency and a second resonant frequency of the ESA and modulating the variable coupling element at a second tonal frequency via a second modulator, wherein the second tonal frequency is a difference between the first resonant frequency and the second resonant frequency of the ESA.

[0008] Additional features and advantages are realized through the techniques of the present disclosure. Other embodiments and aspects of the disclosure are described in detail herein and are considered a part of the claimed disclosure. For a better understanding of the disclosure with the advantages and the features, refer to the description and to the drawings.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0009] For a more complete understanding of this disclosure, reference is now made to the following brief description, taken in connection with the accompanying drawings and detailed description, wherein like reference numerals represent like parts:

[0010] FIG. 1 depicts a system in an illustrative embodiment;

[0011] FIG. 2 is a diagram of the antenna assembly of FIG. 1, in an illustrative embodiment;

[0012] FIG. 3 shows a graph illustrating bandwidth for the antenna assembly when operating in a constant gain-bandwidth product mode of operation;

[0013] FIG. 4 shows a graph illustrating bandwidth for the antenna assembly when operating in a constant bandwidth mode of operation;

[0014] FIG. 5 shows a graph illustrating bandwidth for the antenna assembly when operating in a wide bandwidth mode of operation;

[0015] FIG. 6 shows a graph of antenna gain and saturation power for an illustrative antenna assembly;

[0016] FIG. 7 shows an antenna assembly in an alternative embodiment;

[0017] FIG. 8 shows an antenna assembly in another alternative embodiment;

[0018] FIG. 9 shows an integrated circuit board that includes the antenna assembly;

[0019] FIG. 10 is a multi-frequency modulation circuit for the antenna assembly, in an illustrative embodiment; and

[0020] FIG. 11 is a graph of input return loss for the multi-frequency modulation circuit of FIG. 10.

DETAILED DESCRIPTION

[0021] FIG. 1 depicts a system 100 in an illustrative embodiment. The system 100 can be an aerial vehicle or aerial device, such as an unmanned aerial vehicle, a guided missile, a satellite orbiting the Earth, etc. The system 100 includes a communication device 102 for communicating

(i.e., transmitting and/or receiving) data which can be used for operating the system. The communication device **102** includes an antenna assembly **104** having an electrically small antenna (ESA **106**) and a drive circuit **108** that sends and/or receives the data via the ESA **106**. An ESA **106** is an antenna that is significantly shorter than the wavelength of the signals which are communication over it. The ESA **106** shown in FIG. **1** is in the form of a loop having a radius a and a loop thickness of $2a_0$. However, this antenna design is not meant to be a limitation of the system.

[0022] FIG. **2** is a diagram **200** of the antenna assembly **104** of FIG. **1**, in an illustrative embodiment. The antenna assembly **104** includes the electrically small antenna (ESA **106**) and the drive circuit **108**. The ESA **106** and the drive circuit **108** are electrically connected via a variable coupling element **202**. In an embodiment, the variable coupling element **202** can be a variable capacitor, such as a semiconductor-based variable capacitor. The drive circuit **108** includes a resonator **204** and an RF electrical signal input/output device (I/O device **206**). The resonator **204** can be an RLC circuit, such as the RLC circuit shown in FIG. **1**, with a resistor, capacitor and inductor connected in parallel with each other. The resonator **204** has a first resonant frequency f_r . The ESA **106** has a second resonant frequency f_{ESA} . A matching inductor **216** is coupled to the ESA **106**.

[0023] A first modulator **208** and a second modulator **210** provide signals for controlling operation of the variable coupling element **202**. The first modulator **208** is coupled to the variable coupling element **202** via a first filter **212** and the second modulator is coupled to the variable coupling element **202** via a second filter **214**. The first modulator **208** is shown on the same side of the variable coupling element **202** as the drive circuit **108** and the second modulator **210** is shown on the same side of the variable coupling element **202** as the ESA **106**. However, this is not meant as a limitation of the antenna assembly **104**.

[0024] The first modulator **208** drives the variable coupling element **202** using a first tonal signal. The first tonal signal operates at a first tonal frequency f_s . The first tonal frequency f_s is a summation of the first resonant frequency f_r and the second resonant frequency f_{ESA} , as shown in Eq. (1):

$$f_s = f_r + f_{ESA} \quad \text{Eq. (1)}$$

[0025] The second modulator **210** drives the variable coupling element **202** using a second tonal signal. The second tonal signal operates at a second tonal frequency f_d . The second tonal frequency f_d is a difference of the first resonant frequency f_r and the second resonant frequency f_{ESA} , as shown in Eq. (2):

$$f_d = |f_r - f_{ESA}| \quad \text{Eq. (2)}$$

[0026] The bandwidth and gain of the antenna assembly can be controlled by controlling the amplitude or magnitude of the first tonal signal and the second tonal signal, as shown in FIGS. **3-5**.

[0027] FIG. **3** shows a graph **300** illustrating bandwidth for the antenna assembly **104** when operating in a constant

gain-bandwidth product mode of operation. Frequency is shown in Megahertz (MHz) along the abscissa and transmission power is shown in decibels (dB) along the ordinate axis. The antenna assembly is operated under the condition of Eq. (3):

$$d^2 - g^2 < \frac{k_r k_{ESA}}{4} \quad \text{Eq. (3)}$$

where d is the amplitude of the difference signal (i.e., the second tonal signal), g is the amplitude of the summation signal (i.e., the first tonal signal), k_r is the linewidth of the resonator **204**, and k_{ESA} is the linewidth of the ESA **106**.

[0028] A resonant matching gain curve **302** shows bandwidth for the antenna assembly **104** operated in resonant matching (i.e., by using a static matching network without modulating tones). The maximum transmission is 0 decibels at 300 MHz. The resonant matching gain curve **302** extends from about 298 MHz to about 302 MHz at -10 dB. With d held constant, as g increases, the maximum transmission power can be increased without significantly increasing the bandwidth. For example, at $d=1$ and $g=1.55$, gain curve **304** has a maximum transmission power at about 14 dB at 300 MHz and extends from about 295 MHz to about 305 MHz at -10 dB.

[0029] FIG. **4** shows a graph **400** illustrating bandwidth for the antenna assembly **104** when operating in a constant bandwidth mode of operation. Frequency is shown in Megahertz (MHz) along the abscissa and transmission power is shown in decibels (dB) along the ordinate axis. The antenna assembly is operated under the condition of Eq. (4):

$$d^2 - g^2 = \frac{k_r k_{ESA}}{4} \quad \text{Eq. (4)}$$

The resonant matching gain curve **302** is shown. Both g and d can be adjusted simultaneously. In general, g and d are either increasing together or decreasing together to satisfy Eq. (4). Gain curve **402** shows bandwidth for the antenna assembly operating with $d=3.5$ and $g=3.36$. The maximum transmission power is at about 11 dB at 300 MHz and extends from about 287 MHz to about 313 MHz at -10 dB, which is increased over the bandwidth of the resonant matching gain curve **302**. Furthermore, perfect impedance match (zero return loss) is obtained in this mode of operation.

[0030] FIG. **5** shows a graph **500** illustrating bandwidth for the antenna assembly **104** when operating in a wide bandwidth mode of operation. Frequency is shown in Megahertz (MHz) along the abscissa and transmission power is shown in decibels (dB) along the ordinate axis. The antenna assembly is operated under the condition of Eq. (4):

$$d^2 - g^2 > \frac{k_r k_{ESA}}{4} \quad \text{Eq. (5)}$$

[0031] The resonant matching gain curve **302** is shown. Both g and d can be adjusted simultaneously. In general, g and d are either increasing together or decreasing together to satisfy Eq. (5). Gain curve **502** shows bandwidth for the antenna assembly operating with $d=4.3$ and $g=3.36$. The

maximum transmission power is relatively unchanged (at about 0 dB at 300 MHz). However, the bandwidth has increased and extends from about 283 MHz to about 317 MHz at -10 dB.

[0032] The antenna assembly **104** can be operated in either of the modes of FIGS. 3-5 based on a desired application of the system **100**. For example, the antenna assembly **104** can be operated in the wide bandwidth mode to scan a particular area, as shown in FIG. 5. If there is an object of interest, the antenna assembly can be switched to operate in the constant gain-bandwidth product mode as shown in FIG. 3 to focus on the object while filtering out out-of-band noise.

[0033] FIG. 6 shows a graph **600** of antenna gain for an illustrative antenna assembly. Input power is shown in decibel-meters (dBm) along the abscissa and gain is shown in decibels (dB) along the ordinate axis. The gain curve **602** is generally constant below an input power threshold **604** (e.g., about -4 dBm), above which the signal experiences compression.

[0034] FIG. 7 shows an antenna assembly **700** in an alternative embodiment. The antenna assembly **700** includes a coupling element **702** that controls an electrical connection between a drive circuit **704** and an ESA **706**. The coupling element **702** is a diode rectifier bridge. The drive circuit **704** includes a positive signal input **708** (V_{sig}^+) and a negative signal input **710** (V_{sig}^-).

[0035] The diode rectifier bridge includes four nodes and four sides connecting the four nodes, each side having a diode thereon. The diode(s) can be a varactor (i.e., a diode whose internal capacitance changes with respect to a reverse voltage applied thereto). The nodes include a first signal node **712**, a second signal node **714**, a first modulation node **716** and second modulation node **718**. The positive signal input **708** is supplied to the first signal node **712** and the negative signal input **710** is supplied to the second signal node **714**. The ESA **706** is connected across the first signal node **712** and the second signal node **714**. A first modulator **720** (V_{pump}^+) provides the summation signal (P_s) (i.e., the first tonal signal) into the diode rectifier bridge at the first modulation node **716** and the second modulator **722** (V_{pump}^-) provides the difference signal (P_d) (i.e., the second tonal signal) into the diode rectifier bridge at the first modulation node **716**. The second modulation node **718** leads to ground for the RF signal. A voltage bias can be applied to the diodes (varactors) so that they operate in a reverse biased region. The reverse bias reduces loss and noise in the RF signal and improves signal linearity. In general, a DC voltage bias can be applied across the modulation nodes, with a positive voltage value and the first modulation node **716** and a negative voltage value at the second modulation node **718**.

[0036] Returning to FIG. 6, the saturation power can be increased when the antenna assembly **700** of FIG. 7 is being used by stacking multiple varactors and/or by increasing a voltage bias across one or more varactors.

[0037] FIG. 8 shows an antenna assembly **800** in another alternative embodiment. The antenna assembly **800** includes a coupling element **802** that controls an electrical connection between the drive circuit **804** and an ESA **806**. The coupling element **802** is a variable capacitor rectifier, which is similar to the diode rectifier bridge of FIG. 7 with variable capacitors instead of diodes.

[0038] The variable capacitor rectifier includes a first antenna node **808**, a second antenna node **810**, a first

modulation node **812** and a second modulation node **814**. A first variable capacitor **816a** is between the first antenna node **808** and the first modulation node **812**. A second variable capacitor **816b** is between the first antenna node **808** and the second modulation node **814**. A third variable capacitor **816c** is between the first modulation node **812** and the second antenna node **810**. A fourth variable capacitor **816d** is between the second antenna node **810** and the second modulation node **814**. The first variable capacitor **816a** and the fourth variable capacitor **816d** are operated in phase with each other. The second variable capacitor **816b** and the third variable capacitor **816c** are operated in phase with each other and 180 degrees out of phase with the first variable capacitor **816a** and the fourth variable capacitor **816d**.

[0039] A positive terminal of a signal source **820** is connected to the first modulation node **812** and a negative terminal of the signal source is connected to the second modulation node **814**. The ESA **806** is connected across the first antenna node **808** and the second antenna node **810**. A signal modulator **822** provides a first tonal signal (sum) and a second tonal signal (difference) across the first modulation node **812** and the second modulation node **814**.

[0040] FIG. 9 shows an integrated circuit board **900** that includes the antenna assembly. The drive circuit **902**, ESA **904** and coupling element **906** are elements of the integrated circuit board **900**. The drive circuit **902** includes the positive signal **908** (V_{sig}^+), the negative signal **910** (V_{sig}^-), the first modulator **912** (V_{pump}^+), and the second modulator **914** (V_{pump}^-).

[0041] FIG. 10 is a multi-frequency modulation circuit **1000** for the antenna assembly, in an illustrative embodiment. The multi-frequency modulation circuit **1000** includes an electrically small antenna (ESA **1002**), a drive circuit **1004**, and a modulation circuit or variable coupling element **1006** that electrically couples the drive circuit **1004** to the ESA **1002**.

[0042] The variable coupling element **1006** connects to the driver circuit **1004** to the driver circuit **1004** via a positive voltage line **1008** and a negative voltage line **1010**. The variable coupling element **1006** includes a four variable capacitors. A first variable capacitor **1012** and a third variable capacitor **1016** both extend from the positive voltage line **1008** to the negative voltage line **1010**. A second variable capacitor **1014** extends from the first variable capacitor **1012** to the third variable capacitor **1016** along the positive voltage line **1008**, and a fourth variable capacitor **1018** extends from variable capacitor **1012** to the third variable capacitor **1016** along the negative voltage line **1010**. The second variable capacitor **1014** and the fourth variable capacitor **1018** can have a same capacitance or can be operated in the same manner.

[0043] The first variable capacitor **1012** is operated using a modulation producing a time-evolution of its capacitance is described in Eq. (6):

$$C_1(t) = 2 \frac{a - c(t)}{ad - c^2(t)} \approx \sum c_{1n} \cos(\omega_n t) \quad \text{Eq. (6)}$$

The second variable capacitor **1014** and the fourth variable capacitor **1018** are operated using a modulation producing a time-evolution of their capacitance is described in Eq. (7):

$$C_2(t) = 2 \frac{c(t)}{ad - c^2(t)} \approx \sum c_{2n} \cos(\omega_n t) \quad \text{Eq. (7)}$$

The third variable capacitor **1016** is operated using a modulation producing a time-evolution of its capacitance is described in Eq. (7):

$$C_3(t) = 2 \frac{d - c(t)}{ad - c^2(t)} \approx \sum c_{3n} \cos(\omega_n t) \quad \text{Eq. (8)}$$

Parameters a and d can be controlled or preselected. For Eqs. (6)-(8), the variable c(t) is given as shown in Eq. (9):

$$c(t) = c_0 + 2c_1 \cos(\Omega t + \phi) \quad \text{Eq. (9)}$$

where c_0 can c_1 can be controlled or preselected. The phases ϕ of the capacitors can be separated by 90 degrees. While four capacitors are shown, a variable coupling element having more than four capacitors can also be used to provide multi-frequency modulation.

[0044] FIG. 11 is a graph **1100** of input return loss for the multi-frequency modulation circuit **1000** of FIG. 10. For graph **1100**, the parameters are as given in Eqs. (10)-(13):

$$a = d = 1 \quad \text{Eq. (10)}$$

$$c_0 = a/2 \quad \text{Eq. (11)}$$

$$c_1 = 0.9c_0/2 \quad \text{Eq. (12)}$$

$$\Omega = 0.95 \quad \text{Eq. (13)}$$

The graph **1100** shown a first curve **1102** representing g a Bode Fano-limit and a second curve **1104** representing the input return loss for an antenna assembly operating with the variable coupling element **1006**. The graph **1100** shows that complex modulation is possible with matching that extends beyond the Bode-Fano limit.

[0045] The corresponding structures, materials, acts, and equivalents of all means or step plus function elements in the claims below are intended to include any structure, material, or act for performing the function in combination with other claimed elements as specifically claimed. The description of the present disclosure has been presented for purposes of illustration and description, but is not intended to be exhaustive or limited to the disclosure in the form detailed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the disclosure. The embodiments were chosen and described in order to best explain the principles of the disclosure and the practical application, and to enable others of ordinary skill in the art to understand the various embodiments with various modifications as are suited to the particular use contemplated.

[0046] While the preferred embodiments have been described, it will be understood that those skilled in the art, both now and in the future, may make various improvements and enhancements which fall within the scope of the claims

which follow. These claims should be construed to maintain the proper protection for the disclosure as first described.

What is claimed is:

1. An antenna assembly, comprising:

an electrically small antenna (ESA);

a drive circuit for operating the ESA, the drive circuit including a resonator having a first resonant frequency;

a variable coupling element that controls an electrical connection between the drive circuit and the ESA;

a first modulator configured to modulate the variable coupling element at a first tonal frequency, wherein the first tonal frequency is a sum of the first resonant frequency and a second resonant frequency of the ESA; and

a second modulator configured to modulate the variable coupling element at a second tonal frequency, wherein the second tonal frequency is a difference between the first resonant frequency and the second resonant frequency.

2. The antenna assembly of claim **1**, wherein the variable coupling element is modulated simultaneously by the first modulator and the second modulator.

3. The antenna assembly of claim **1**, wherein the resonator is an RLC circuit and the first resonant frequency is a resonant frequency of the RLC circuit.

4. The antenna assembly of claim **3**, wherein the drive circuit further includes an input/output device in parallel with the resonator.

5. The antenna assembly of claim **1**, wherein the variable coupling element is one of: (i) a variable capacitor; and (ii) a varactor.

6. The antenna assembly of claim **1**, wherein the variable coupling element is one of: (i) a diode rectifier bridge; and (ii) and variable capacitor rectifier bridge.

7. The antenna assembly of claim **1**, wherein the drive circuit, the ESA, the variable coupling element, the first modulator, and the second modulator are elements of an integrated circuit board.

8. The antenna assembly of claim **1**, wherein the variable coupling element further comprises four variable capacitors.

9. A device, comprising:

an electrically small antenna (ESA);

a drive circuit for operating the ESA, the drive circuit including a resonator having a first resonant frequency;

a variable coupling element that controls an electrical connection between the drive circuit and the ESA;

a first modulator configured to modulate the variable coupling element at a first tonal frequency, wherein the first tonal frequency is a sum of the first resonant frequency and a second resonant frequency of the ESA; and

a second modulator configured to modulate the variable coupling element at a second tonal frequency, wherein the second tonal frequency is a difference between the first resonant frequency and the second resonant frequency.

10. The device of claim **9**, wherein the variable coupling element is modulated simultaneously by the first modulator and the second modulator.

11. The device of claim **9**, wherein the resonator is an RLC circuit and the first resonant frequency is a resonant frequency of the RLC circuit.

12. The device of claim **9**, wherein the variable coupling element is one of: (i) a variable capacitor; and (ii) a varactor.

13. The device of claim **9**, wherein the variable coupling element is one of: (i) a diode rectifier bridge; and (ii) a variable capacitor rectifier bridge.

14. The device of claim **9**, wherein the drive circuit, the ESA, the variable coupling element, the first modulator, and the second modulator are elements of an integrated circuit board.

15. The device of claim **9**, wherein the device is one of: (i) an aerial device; (ii) an unmanned aerial vehicle; (iii) a guided missile; and (iv) a satellite orbiting the earth.

16. A method of operating an electrically small antenna, comprising:

controlling an electrical connection between the electrically small antenna and a drive circuit via a variable coupling element, the drive circuit including a resonator having a first resonant frequency, wherein controlling the electrical connection comprises:

modulating the variable coupling element at a first tonal frequency via a first modulator, wherein the first tonal frequency is a sum of the first resonant frequency and a second resonant frequency of the ESA; and

modulating the variable coupling element at a second tonal frequency via a second modulator, wherein the second tonal frequency is a difference the first resonant frequency and the second resonant frequency of the ESA.

17. The method of claim **16**, further comprising modulating the variable coupling element simultaneously at the first tonal frequency and the second tonal frequency.

18. The method of claim **16**, wherein the resonator is an RLC circuit and the first resonant frequency is a resonant frequency of the RLC circuit.

19. The method of claim **16**, wherein the variable coupling element is one of: (i) a diode rectifier bridge; and (ii) a variable capacitor rectifier bridge.

20. The method of claim **1**, wherein the variable coupling element includes four capacitors, further comprising controlling the variable coupling element using three or more independent modulations signals.

* * * * *